OR4023 (Optimization) Quiz 1

Name:_____ Student ID:_____

Instructions:

1. No calculator is allowed.
2. Duration of quiz: 90 minutes

Question 1. (10 marks) Using only definitions to prove

a) The closed ball $B=\{\boldsymbol{x}\in\mathbb{R}^n:\|x\|\leqslant r\}$ is convex.

b) The function $f(x)=|x|$ is convex over $\mathbb{R}$.

Answer.

a) Let $\boldsymbol{x},\boldsymbol{y}\in B$ and $\theta\in(0,1)$. Then

$$\|\theta\boldsymbol{x}+(1-\theta)\boldsymbol{y}\|\leqslant\theta\|\boldsymbol{x}\|+(1-\theta)\|\boldsymbol{y}\|\leqslant\theta r+(1-\theta)r=r\Longrightarrow\theta\boldsymbol{x}+(1-\theta)\boldsymbol{y}\in B.$$

Thus B is convex by definition.

b) Let $x,y\in\mathbb{R}$ and $\theta\in(0,1)$. Then

$$f(\theta x+(1-\theta)y)=|\theta x+(1-\theta)y|\leqslant\theta|x|+(1-\theta)|y|=\theta f(x)+(1-\theta)f(y).$$

Hence f is convex by definition.

Question 2. (30 marks) Consider the function $f(x)=\frac{1}{3}x^3+\frac{1}{2}x^2-x$.

a) Show that f has a unique local minimum in (0,1), which is a strict local minimum.

b) Apply two iterations of the bisection method to find an approximate value of the local minimum (use the midpoint in the final interval of uncertainty as the approximate solution).

c) Give an error estimate for the approximate solution you find in (b) without using the exact value of the minimum.

d) Apply one iteration of the classical Newton's method with the initial guess $x_0=0.5$ to find an approximate value of the local minimum.

Answer.

a) We have $f'(x)=x^2+x-1$ and $f'(0)<0, f'(1)>0$. Hence there exists at least one $x^*\in(0,1)$ such that $f'(x^*)=0$. Since $f''(x)=2x+1>0$ for all $x\in(0,1)$, there is only one such x^*, which is a strict local minimizer (also a global one).

b) Let $g(x)=f'(x)=x^2+x-1$. Then $g(0)=-1, g(1)=1$.
First iteration: Let

$$a=0, b=1, c=0.5.$$

Then $g(0.5)=-0.25$. $g(0.5)$ and $g(1)$ have different signs, then $x^*\in[0.5,1]$.
Second iteration: Let

$$a=0.5, b=1, c=0.75.$$

Then $g(0.75)=0.3125$. $g(0.5)$ and $g(0.75)$ have different signs, then $x^*\in[0.5,0.75]$.
Hence [0.5,0.75] is the final interval of uncertainty. The midpoint is

$$x=0.625.$$

c) We know the exact minimizer x^* lies in [0.5,0.75] and x=0.625 is the midpoint. Error estimate:

$$|x-x^*|\leqslant 0.125.$$

d)

$$x_1=x_0-\frac{f'(x_0)}{f''(x_0)}=0.5-\frac{-0.25}{2}=0.625.$$

Question 3. (20 marks) Consider the function $f(x_1,x_2)=x_1^2+2x_2^2-2x_1-4x_2$.

a) Find a local minimum point $\mathbf{x}^*$ of f. Is it also a global minimum? Is it unique? Explain your answers.

b) Starting from the initial guess $\mathbf{x}^{(0)}=(0,0)$, find the next point $\mathbf{x}^{(1)}$ by the steepest descent method with exact line search.

Answer. We have

$$\nabla f(x_1,x_2)=[2x_1-2,4x_2-4],\quad \nabla^2 f(x_1,x_2)=\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}.$$

a) Stationary point:

$$\nabla f(x_1,x_2)=0\Rightarrow \mathbf{x}^*=[1,1]^T.$$

Since $\nabla^2 f(x_1,x_2)$ is SPD everywhere, we know $\mathbf{x}^*$ is a local minimum, which is also a global minimum and the unique minimum.

b) The search direction is

$$\mathbf{p}_0=-\nabla f(\mathbf{x}^{(0)})=[2,4]^T.$$

Let

$$\varphi(\alpha)=f(\mathbf{x}^{(0)}+\alpha\mathbf{p}_0)=f([2\alpha,4\alpha]^T)=36\alpha^2-20\alpha.$$

The minimum of $\varphi(\alpha)$ is $\alpha=\frac{5}{18}$. Thus

$$\mathbf{x}_1=\mathbf{x}^{(0)}+\alpha\mathbf{p}_0=\left[\frac{5}{9},\frac{10}{9}\right]^T.$$

Question 4. (20 marks) Use optimization to find the minimal distance between the point (1,1,1) and the plane $z=1-x-y$. *Hint: Formulate an optimization problem* $\min_{(x,y)\in\mathbb{R}^2} f(x,y)$.

Answer. The square of the distance is given by

$$f(x,y)=(x-1)^2+(y-1)^2+(z-1)^2=(x-1)^2+(y-1)^2+(x+y)^2=2(x^2+y^2+xy-x-y+1).$$

Remark. One can also define the square root of the above as the function, but the above is easier to compute.

Consider the optimization problem $\min_{(x,y)\in\mathbb{R}} f(x,y)$. We have

$$\nabla f=2[2x+y-1,2y+x-1]=0\Longrightarrow x=y=\frac{1}{3}.$$

Moreover, the Hessian

$$\nabla^2 f=2\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

is SPD everywhere. Hence $\left(\frac{1}{3},\frac{1}{3}\right)$ is the global minimum point. The minimal distance is

$$\sqrt{f\left(\frac{1}{3},\frac{1}{3}\right)}=\frac{2}{\sqrt{3}}.$$

Question 5. (20 marks) Fit the data points (0,1),(1,2),(2,4) by a linear function $y=ax+b$ using the least squares method. Sketch a graph to illustrate your answer.

Answer. Let

$$X=\begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}, \quad Y=\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}.$$

Then

$$X^TX=\begin{pmatrix} 5 & 3 \\ 3 & 3 \end{pmatrix}, \quad X^TY=\begin{pmatrix} 10 \\ 7 \end{pmatrix}.$$

Using the Cramer's rule, we have

$$\begin{pmatrix} a \\ b \end{pmatrix}=(X^TX)^{-1}(X^TY)=\left(\frac{3}{2},\frac{5}{6}\right).$$